Report* on Experiments:
Enzyme-Linked Immunosorbent Assay, ELISA

https://github.com/Revewill/Biochemistry.github.io/tree/main/Experiments/Report1

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** This is a report for ELISA in Basic Practical Biochemistry.*

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Enzyme-linked immunosorbent assay (ELISA) combines specific antigen-antibody (Ag-Ab) binding with enzyme-catalyzed color development to detect and quantify a target. In indirect ELISA, primary antibodies (1°Ab) bound to an immobilized antigen are detected by an enzyme-conjugated secondary antibody (2°Ab). The resulting absorbance provides a measure of antibody content within a calibrated range.

In this experiment, rabbit anti-human immunoglobulin (IgG) was measured by indirect ELISA with peroxidase (POD)-conjugated goat anti-rabbit IgG, developing colored products with a maximum absorption at 490 nm. A serial dilution series was used to construct a nonlinear calibration curve and estimate the relative antibody content of sample A, prepared at a dilution of 1:2000. Mean recovery was 83.9%, indicating underestimation relative to the nominal value. Variation among replicate wells and the effect of blocking were also assessed as insignificant. Uneven pipetting and bubbles observed during the experiment may have contributed to certain measurement variability. The assay detected dilution-dependent differences in antibody content, although incomplete recovery and replicate variation limited quantitative accuracy and precision.

Indirect ELISA detects antigen-specific antibodies through sequential binding to an immobilized Ag, followed by an enzyme-conjugated secondary antibody (2°Ab).[1] Here, human IgG as Ag was adsorbed onto a microplate to capture rabbit anti-human IgG (1°Ab). Blocking was used to occupy remaining adsorption sites and reduce nonspecific binding. After incubation with 1°Ab, POD-conjugated goat anti-rabbit IgG (2°Ab) was added to detect the bound 1°Ab. Washing between steps with phosphate-buffered saline with Tween-20 (PBS-T) removed unbound reagents. The conjugated POD then catalyzed the oxidation of o-phenylenediamine (OPD) by H_2O_2, producing a colored product measured at 490 nm after termination. Absorbance reflects the amount of 2°Ab retained in the wells and therefore signals 1°Ab binding. This relationship is nonlinear, since binding capacity and color development can affect the response. See Fig.1a for conceptual visualization of the procedure.

The experiment was designed to evaluate quantitative performance under our assay conditions and assess the consistency of our experimental technique. Six successive twofold 1°Ab dilutions, from 1:400 to 1:12800, were established as calibration range. Sample A, prepared at a known dilution of 1:2000, provided an unknown sample and enabled assessment on calibration results. Each condition was measured in three repetitive wells. No-1°Ab controls supplied the background correction, while no-Ag controls assessed blank signals without Ag coating. An unblocked condition was prepared and compared with the corresponding blocked condition to examine how blocking affected the total signal.

During calibration, a total of five nonlinear models were compared, and a power model was selected by cross-validation, giving an R^2 of 0.9711. Sample A showed a mean recovery of 83.9%, indicating that the assay underestimated its nominal antibody content. Within-condition absorbance coefficients of variation (CVs) ranged from 3.4% to 12.7%. Their differences were not statistically significant, indicating an acceptable parallelism. Blocked and unblocked wells likewise showed no significant difference in total signal. Uneven delivery from the multichannel pipette and bubbles observed during measurement may have contributed to certain variations and errors. These observations provide a basis for evaluating both the calibration and the practical limitations of the assay.

Experimental procedures are provided in the Methods section. Data were processed and plotted in R, with AI assistance in writing and revising the scripts. All code is available in the accompanying GitHub repository (https://github.com/Revewill/Biochemistry.github.io/tree/main/Experiments/Report1). The report also discusses what the experiment shows about error analysis.

Calibration and residual analysis

Raw absorbance readings at 490 nm (A_{490}) were assigned to their experimental conditions according to the plate layout (Fig.1b; layout articulated in Methods). The no-1°Ab controls gave a mean background absorbance of 0.0417, which was subtracted from all standard and sample A readings. No-Ag controls were not additionally subtracted.

Six standard dilutions, each measured in triplicate, were used for calibration. Relative 1°Ab content was expressed as the reciprocal of the dilution denominator. Five nonlinear models, namely logarithmic, hyperbolic, power, exponential saturation and four-parameter logistic models, were compared (Table 1). Predictive performance was assessed by leaving out one dilution level at a time, fitting the remaining levels and predicting the omitted readings.

The power model gave the lowest cross-validation prediction error among the models with stable fits and was selected for the calibration (Fig.1c), giving an R^2 of 0.9711. The equation of the power model is expressed as

$$y = a + bx^q$$

Where y denotes the corrected A_{490} values and x denotes the relative 1°Ab contents represented as the reciprocal of the corresponding dilution denominators. The model gave estimated parameters of

$$a = -0.266136135482723,$$
$$b = 5.66238209856632,$$
$$q = 0.268876072140957.$$

The association between relative antibody content and mean absorbance was significant across standard levels ($P = 0.0028$).

Residuals were calculated as observed minus fitted A_{490} (Fig.1d). Both positive and negative residuals occurred across the calibration range, spreading among wells at several dilution levels. The fitted curve is therefore qualified to account for the overall response.

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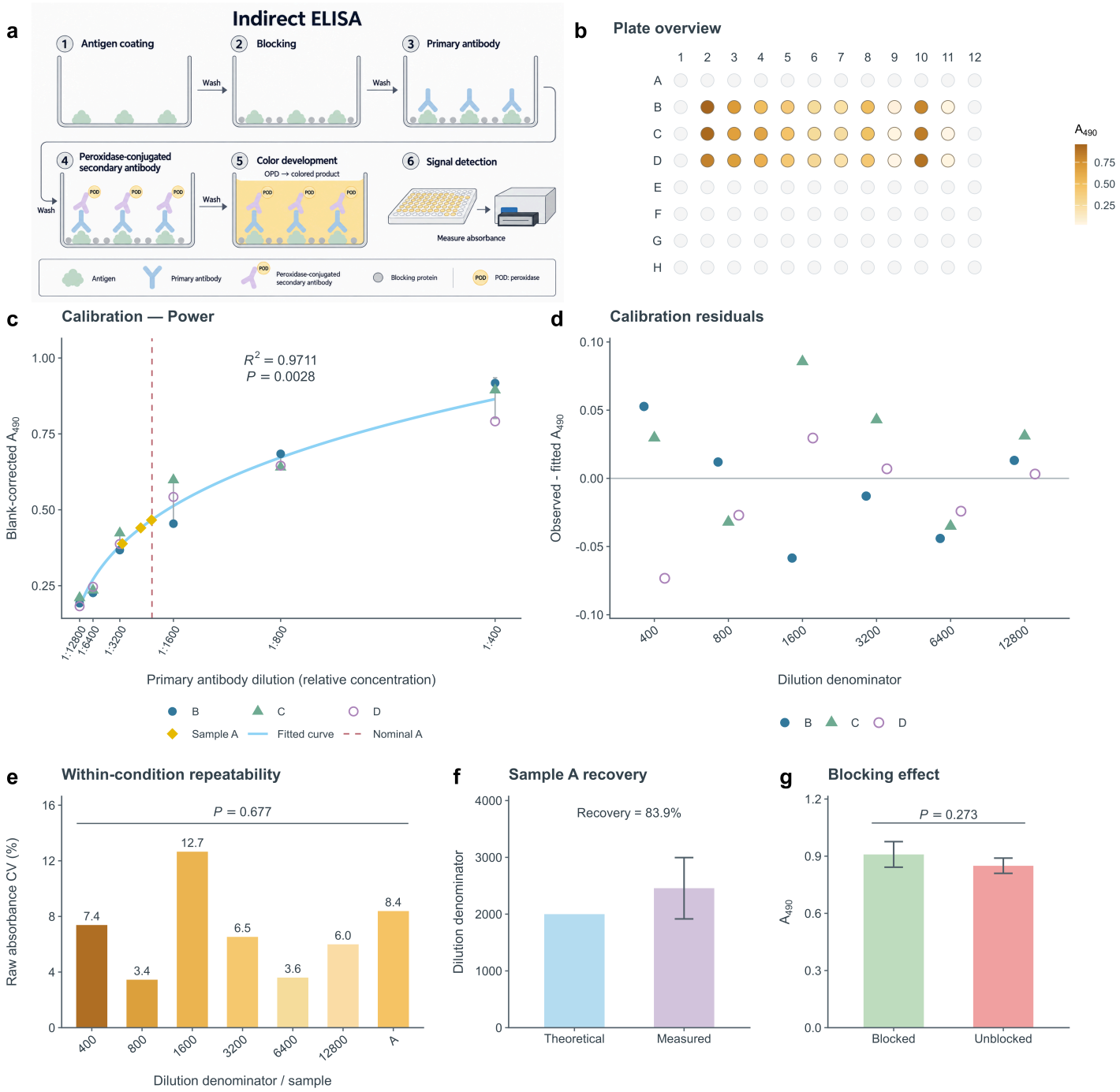


Fig. 1 | Indirect ELISA calibration and assessment of assay performance. a, Schematic of indirect ELISA procedure. **b**, Plate map showing raw A_{490} values. **c**, Background-corrected calibration fitted with the selected power model. Blue circles, green triangles and purple open circles represent wells in rows B, C and D, respectively; yellow diamonds indicate back-calculated sample A values. The dashed line marks the nominal dilution of 1:2000. The horizontal axis represents relative antibody content (the reciprocal of dilution denominator), labelled as dilution ratios. **d**, Calibration residuals, calculated as observed minus fitted A_{490}. **e**, Within-condition coefficients of variation (CVs), calculated from raw absorbance; bar colors indicate group mean absorbance. **f**, Nominal and estimated dilution denominators for sample A. **g**, Raw absorbance in blocked and unblocked wells at a 1°Ab dilution of 1:400. Each condition comprised a triplet; error bars in **c**, **f** and **g** show mean ± SD, with no uncertainty assigned to the nominal value in **f**. Statistical analyses were performed using two-sided Spearman rank correlation across six standard-level means (**c**), global equality-of-CV likelihood-ratio test calibrated by parametric bootstrap (**e**) and two-sided Welch's *t*-test (**g**).

Repeatability across assay conditions

Within-condition repeatability was evaluated using the coefficient of variation (CV), calculated as the sample standard deviation (SD) divided by the mean raw A_{490} and expressed as a percentage.

$$\mathrm{CV} = \frac{\mathrm{SD}}{\mathrm{mean}} \times 100\%$$

Here we used raw readings because background subtraction changes the mean and, consequently, CV, while leaving the within-condition SD unchanged.

The six standard groups had CVs ranging from 3.4% to 12.7%, while sample A had a CV of 8.4% (Fig.1e). The largest variation occurred at the 1:1600 dilution. An exploratory test of equal population CVs, using a normal-model likelihood-ratio statistic with parametric bootstrap calibration, detected no significant difference among the seven conditions ($P = 0.677$). Uneven pipetting and bubbles are possible contributors to the observed spread.

Table 1 | Comparison of nonlinear calibration models for indirect ELISA.

Model	LOCO-RMSE	AICc	R^2	Validation stability
Power	0.0535	−104.362	0.971081	TRUE
Hyperbola	0.0642	−103.091	0.968965	TRUE
Log	0.0782	−99.056	0.953192	TRUE
Exponential saturation	0.0887	−100.483	0.964126	TRUE
4PL	NA	−100.529	0.971226	FALSE

Models are ranked by leave-one-concentration-level-out cross-validation root mean squared error (LOCO-RMSE), with lower values indicating better predictive performance. NA denotes an unavailable LOCO-RMSE owing to unstable validation fits. AICc represents the small-sample corrected Akaike information criterion; R^2 describes the fit to all 18 standard readings; validation stability indicates whether fits remained stable across all omitted levels. 4PL denotes the four-parameter logistic model. The power model was selected for sample back-calculation. Values have been rounded in this table.

Recovery of sample A

Sample A was prepared at a known dilution of 1:2000 by mixing 200 µL of 1:400 1°Ab with 800 µL PBS-T, and was treated as unknown during back-calculation. Each corrected reading was input into the selected calibration model to estimate its relative antibody content. All three readings fell within the standard response range, allowing interpolation.

Recovery was calculated for each well as the estimated content divided by the nominal content in percentages. Mean recovery was 83.9%, indicating an underestimation of 16.1%. Fig.1f compares the nominal dilution denominator with the mean ± SD of the estimated denominators. Data are included in Table 2.

The incomplete recovery indicates a discrepancy between the prepared sample and its estimate. Sample preparation, variation in well responses and calibration uncertainty may all contribute.

Effect of blocking on assay signal

Blocking was evaluated by comparing blocked and unblocked wells (2B-D and 10B-D on 96-well plate, respectively) at the same 1°Ab dilution of 1:400. Mean raw absorbance was 0.909 in blocked wells and 0.850 in unblocked wells, a decrease of 6.5% (Fig.1g). Bars show mean ± SD.

Two-sided Welch's *t*-test detected no significant difference between blocked and unblocked conditions ($P = 0.273$). This comparison measures the effect of blocking on total assay signal, which includes both specific and nonspecific contributions. Therefore, it does not demonstrate that blocking was unnecessary.

Table 2 | Back-calculated antibody content and recovery of sample A.

Well	Corrected A_{490}	Estimated content	Estimated denominator	Recovery (%)
8B	0.4663	0.0004973	2011.02	99.4520
8C	0.4403	0.0004347	2300.35	86.9434
8D	0.3883	0.0003271	3056.93	65.4252

Corrected A_{490} denotes absorbance after subtraction of the mean no-1°Ab control. Estimated content is the relative antibody fraction back-calculated using the selected power model; its reciprocal gives the estimated dilution denominator. Recovery was calculated as estimated content divided by the nominal fraction (1/2000), multiplied by 100%. 8B-D are three replicates of sample A. Calculations used unrounded values. The table used rounded ones for presentation.

Discussion

Indirect ELISA produced a dilution-dependent response that allowed sample A to be evaluated against a standard series. The selected power model described the calibration data with an R^2 of 0.9711, while mean recovery was 83.9%. Thus, the assay detected differences in relative antibody content but underestimated the nominal content of sample A. Neither the comparison of within-condition CVs nor the blocked-unblocked comparison reached statistical significance. However, significance should only be interpreted within limited scope of three replicates per condition.

A strength of this experiment was the evaluation of several aspects of measurement quality rather than reliance on the calibration curve alone. Model comparison, residual analysis, replicate CVs and recovery provided complementary information about predictive performance and experimental variation. Cross-validation supported model selection without using sample A, preserving its role as a separate recovery check. Processing the data in R also retained a traceable record of background correction, fitting and back-calculation. During lab work, advance planning and coordination allowed the procedure to proceed efficiently, with little unnecessary delay between steps.

The most evident procedural problem, however, was persistent foam during washing. After washing solution was discarded by flicking, substantial foam remained in some wells and was still present when the plate was placed in the microplate reader. Attempts to break the bubbles with a fine pipette tip removed only a small portion of the foam. Bubbles in the optical path may alter light transmission and increase variation in A_{490} readings. However, their contribution cannot be quantified from the available data, and they cannot be assumed to account for every observation in this experiment.

A second concern was visibly unequal liquid uptake among the three multichannel pipette channels used for rows B-D. The channel corresponding to row D repeatedly appeared to draw slightly less liquid and also produced less foam. This observation suggests a possible difference in tip sealing or channel performance. Lower delivery could have affected reagent amounts or dilution ratios. Sample A in row D gave the lowest recovery, but the standard readings did not show a uniformly lower response in the row.

Future experiments should include pre-checks of tip sealing and delivery consistency. Washing and liquid removal should also be performed cautiously, with attention to residual liquid and foam. Recording these issues during the experiment proves useful because it identified specific areas for improvement in subsequent experiments.

Overall, indirect ELISA offered a convenient way to compare antibody contents and controls within one plate. Its quantitative performance, nevertheless, depended heavily on consistent liquid handling, washing and signal development, as well as an appropriate calibration model. Nowadays, more sensitive approaches are available when low target abundance is the main limitation. For example, digital ELISA detects individual enzyme-labelled immunocomplexes in small reaction compartments and has achieved substantially lower detection limits than conventional ELISA in protein assays.[2] For the present experiment, however, improving liquid handling and control of bubbles would be a reasonable response to the observed limitations.

References

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Methods

Indirect ELISA

Human IgG was diluted to 0.025 mg/mL (corresponding to a dilution of 1:400) in carbonate-bicarbonate coating buffer (pH 9.6). A 96-well microplate was coated with 200 µL per well in rows B-D, columns 2-10, and incubated at 37°C for 40 min. The coating solution was discarded, and the wells were washed three times with PBS-T (200 µL per well per wash). Blocking solution containing 0.02% (w/v) non-fat dry milk in PBS-T was added at 200 µL per well to columns 2-11, except column 10. The plate was incubated at room temperature for 6 min and washed as described above.

Rabbit anti-human IgG was serially diluted twofold in PBS-T to obtain six dilutions from 1:400 to 1:12800, which were added to columns 2-7, respectively. The dilution series were prepared on a multi-well glass plate by adding 1400 µL 1:400 1°Ab solution to the first well and serial diluting in six 700 µL wells. Sample A, prepared at a dilution of 1:2000 by mixing 200 µL 1:400 1°Ab solution with 800 µL PBS-T, was added to column 8. Column 9 served as the no-1°Ab control. Columns 10 and 11 received 1°Ab at 1:400 and served as the unblocked and no-Ag controls, respectively. Each condition occupied three technical replicates in rows B-D, with a reaction volume of 200 µL per well. After incubation at 37°C for 63 min, the wells were washed as described above.

POD-conjugated goat anti-rabbit IgG was diluted to 1:30000 in PBS-T and added to all assay wells at 200 µL per well. The plate was incubated at 37 °C for 30 min and washed three times with PBS-T. A premixed substrate solution containing OPD and H_2O_2 in citrate-phosphate buffer was supplied ready for use and added at 200 µL per well. After 4 min, color development was stopped by adding 50 µL of 2 M H_2SO_4 per well. Absorbance was measured at 490 nm using a microplate reader.